

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import (
    accuracy_score, precision_score, recall_score, f1_score,
    roc_auc_score, confusion_matrix, classification_report
)
```

```
from google.colab import files
uploaded = files.upload()

import pandas as pd
import io

df = pd.read_csv('listings_ready_change_01.csv')
df.head()
```

Choose Files listings_rea...nge_01.csv

listings_ready_change_01.csv(text/csv) - 2535381 bytes, last modified: 12/2/2025 - 100% done
Saving listings_ready_change_01.csv to listings_ready_change_01 (1).csv

	listing_id	host_id	host_name	host_since	host_response_rate	host_acceptance_rate	host_is_superhost	host_experience	prop
0	2708.0	3008.0	Chas.	16/09/2008 0:00	1.0	1.00	t	17.0	Priv
1	2732.0	3041.0	Louise-Diane	17/09/2008 0:00	1.0	0.50	f	17.0	Priv
2	6931.0	3008.0	Chas.	16/09/2008 0:00	1.0	1.00	t	17.0	Priv
3	7874.0	21700.0	Henry	15/06/2009 0:00	1.0	0.98	t	16.0	Priv
4	8021.0	22460.0	Sherree Strong	20/06/2009 0:00	1.0	1.00	f	16.0	Priv

5 rows × 27 columns

```
# 2) CHỌN FEATURES CHO MÔ HÌNH
# =====
features = [
    "review_scores_rating",
    "host_response_rate",
    "host_acceptance_rate",
    "reviews_per_month",
    "host_experience",
    "number_of_reviews"
]

df = df.dropna(subset=features + ["host_is_superhost"])

X = df[features]
y = df["host_is_superhost"]
```

```
from sklearn.model_selection import train_test_split

# Create X and y from the DataFrame
X = df[features]
y = df["host_is_superhost"].apply(lambda x: 1 if x == 't' else 0)

# Chia tập 80% train - 20% test
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

print("Train size:", X_train.shape)
print("Test size:", X_test.shape)
```

Train size: (7288, 6)
Test size: (1823, 6)

```
def preprocess_features(X, y, features):
    # Điền giá trị thiếu bằng median
    imputer = SimpleImputer(strategy='median')
    X = pd.DataFrame(imputer.fit_transform(X), columns=features)
```

```
from sklearn.model_selection import train_test_split

# X = dữ liệu đầu vào (các feature)
# y = nhãn host_is_superuser (0/1)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
```

```
from sklearn.linear_model import LogisticRegression

model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
```

```
▼ LogisticRegression ⓘ ?
LogisticRegression(max_iter=1000)
```

```
# 5) PREDICT
# =====
y_pred = model.predict(X_test)
y_prob = model.predict_proba(X_test)[: , 1]
```

```
from sklearn.metrics import (
    accuracy_score, precision_score, recall_score,
    f1_score, roc_auc_score, confusion_matrix, classification_report
)

# Tính các chỉ số đánh giá
acc = accuracy_score(y_test, y_pred)
pre = precision_score(y_test, y_pred)
rec = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)
auc = roc_auc_score(y_test, y_prob) # Không dùng pos_label
cm = confusion_matrix(y_test, y_pred)

# In kết quả
print("\n🔴 KẾT QUẢ ĐÁNH GIÁ MÔ HÌNH LOGISTIC REGRESSION")
print("Accuracy: ", round(acc, 4))
print("Precision:", round(pre, 4))
print("Recall:   ", round(rec, 4))
print("F1-score: ", round(f1, 4))
print("ROC-AUC:  ", round(auc, 4))

print("\n🔴 Confusion Matrix:")
print(cm)

print("\n🔴 Classification Report:")
print(classification_report(y_test, y_pred))
```

```
🔴 KẾT QUẢ ĐÁNH GIÁ MÔ HÌNH LOGISTIC REGRESSION
Accuracy:  0.7049
Precision:  0.7064
Recall:     0.8703
F1-score:   0.7799
ROC-AUC:    0.7485

🔴 Confusion Matrix:
[[332 396]
 [142 953]]

🔴 Classification Report:
              precision    recall  f1-score   support

0             0.70       0.46       0.55       728
```

	1	0.71	0.87	0.78	1095
accuracy				0.70	1823
macro avg		0.70	0.66	0.67	1823
weighted avg		0.70	0.70	0.69	1823

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# Lấy trọng số tuyệt đối
coef = np.abs(model.coef_[0])

# Tạo Series để dễ sắp xếp
importance = pd.Series(coef, index=features).sort_values(ascending=False)

# Plot
plt.figure(figsize=(10, 5))
importance.plot(kind='bar', color='royalblue')

plt.title("Feature Importance - Logistic Regression", fontsize=14)
plt.ylabel("Mức độ ảnh hưởng (|coef|)", fontsize=12)
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()

print("\n 🚀 Bảng mức độ ảnh hưởng:")
print(importance)
```



🚀 Bảng mức độ ảnh hưởng:

```
host_response_rate    3.423389
review_scores_rating  3.172314
host_acceptance_rate  1.818986
reviews_per_month     0.154080
host_experience        0.049497
number_of_reviews     0.001398
dtype: float64
```

